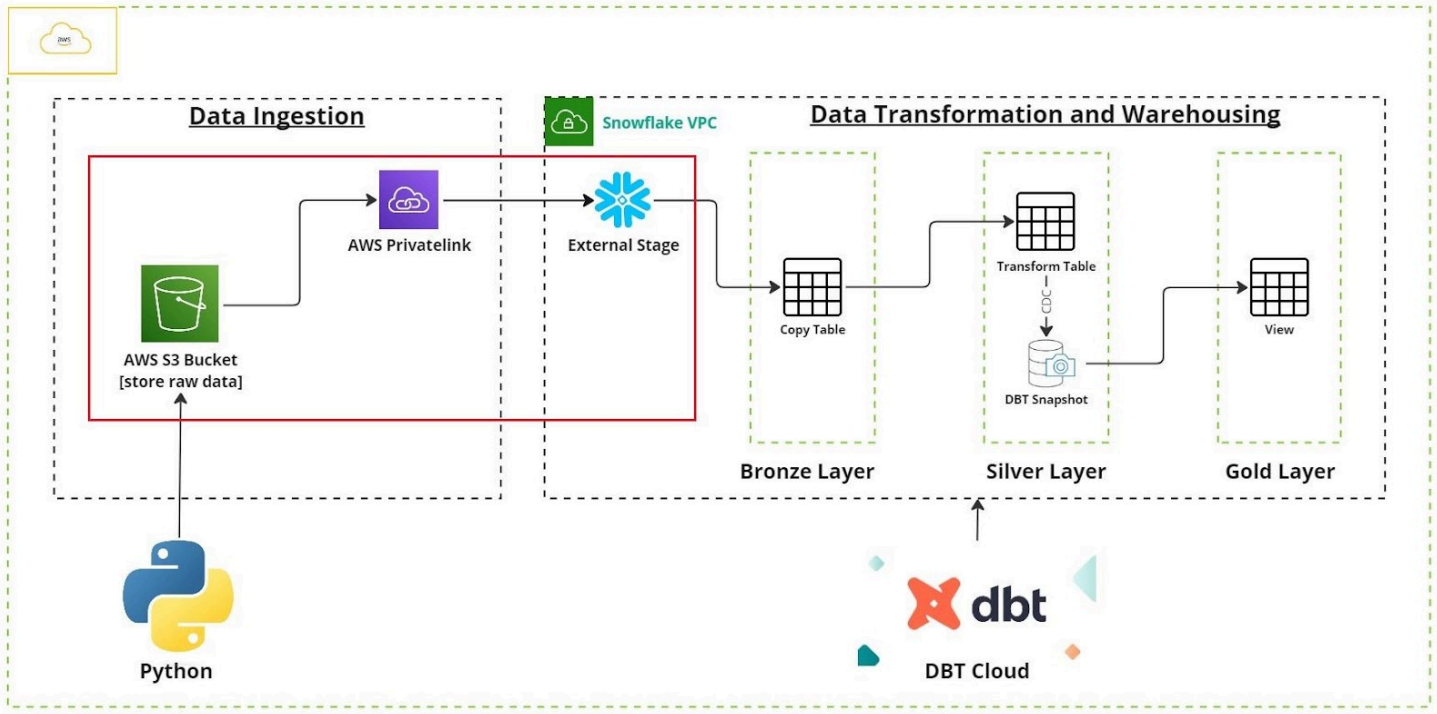
Technical Overview

In this demonstration, we are going to setup S3 Bucket in AWS Console. Along with setup, integration with DynamoDB will also be done.

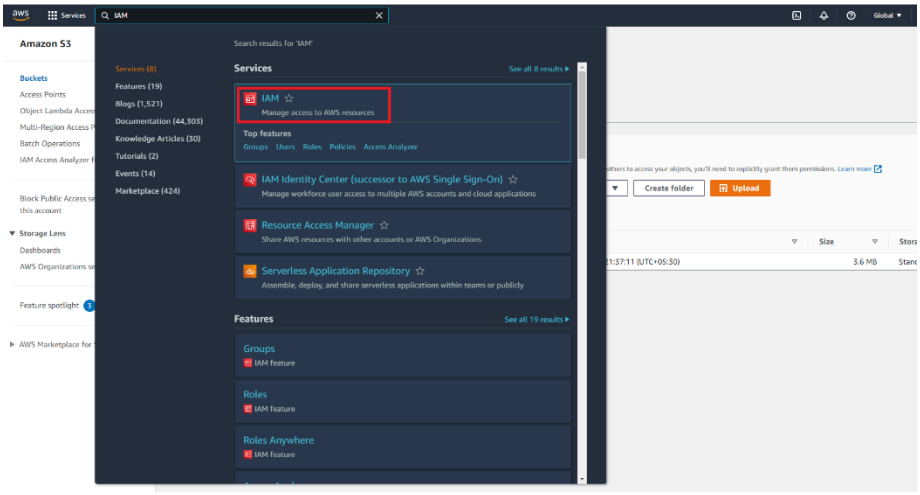


Pre-requisites:

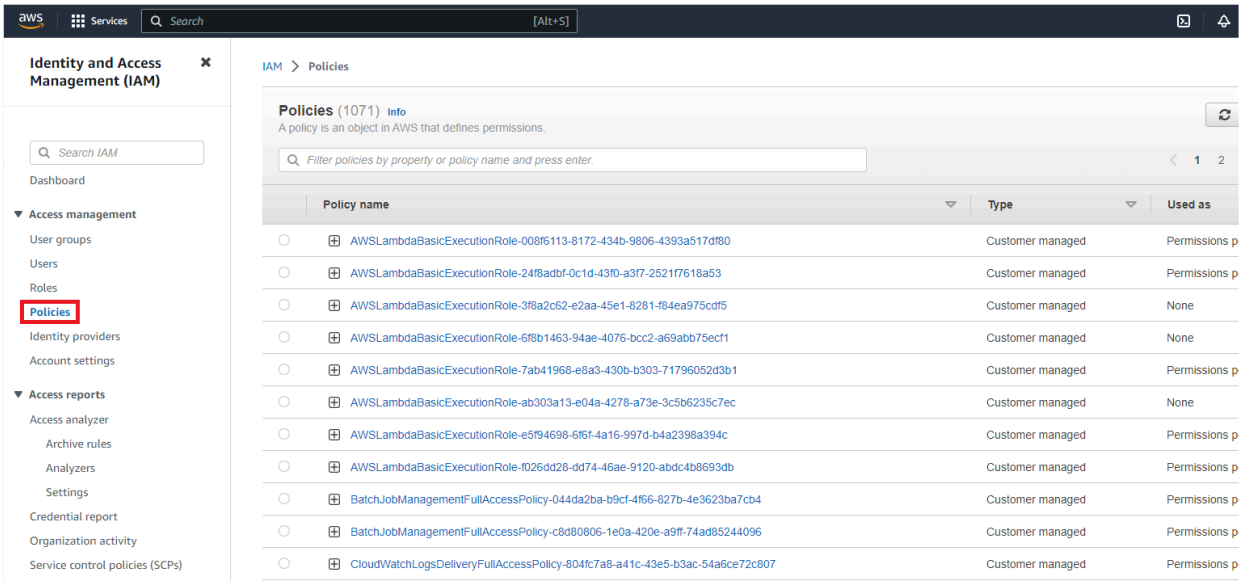
- **AWS Account**
- **Snowflake Account**

Create IAM Policy

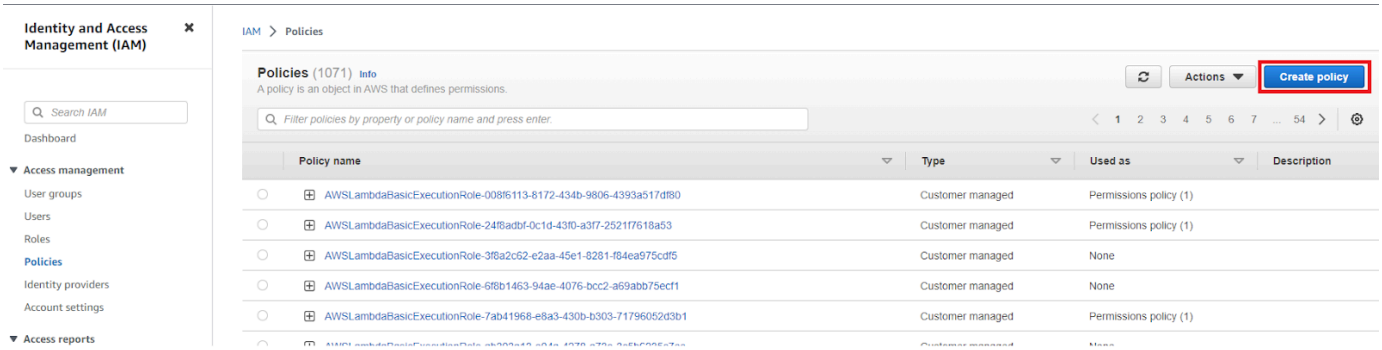
1. **Log into the AWS Management Console.**
2. **From the home dashboard, choose Identity & Access Management (IAM).**



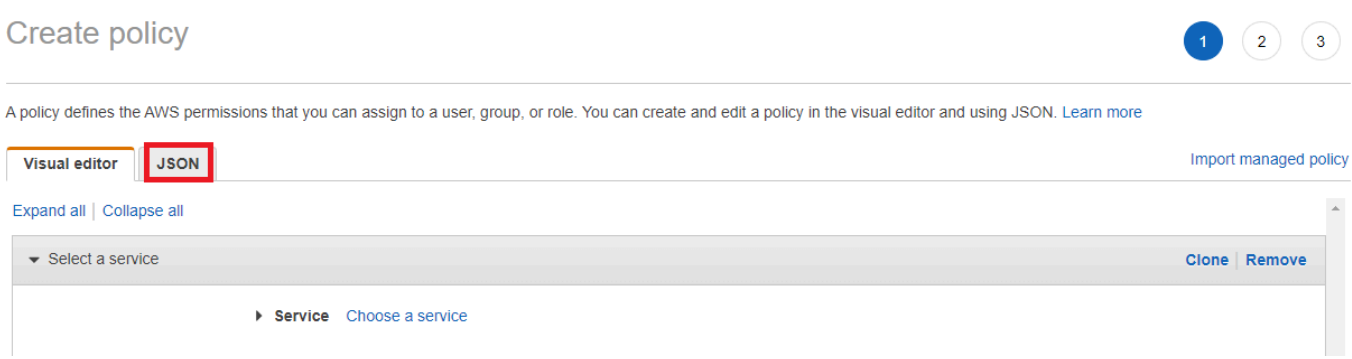
3. Choose Policies from the left-hand navigation pane.



4. Click Create Policy



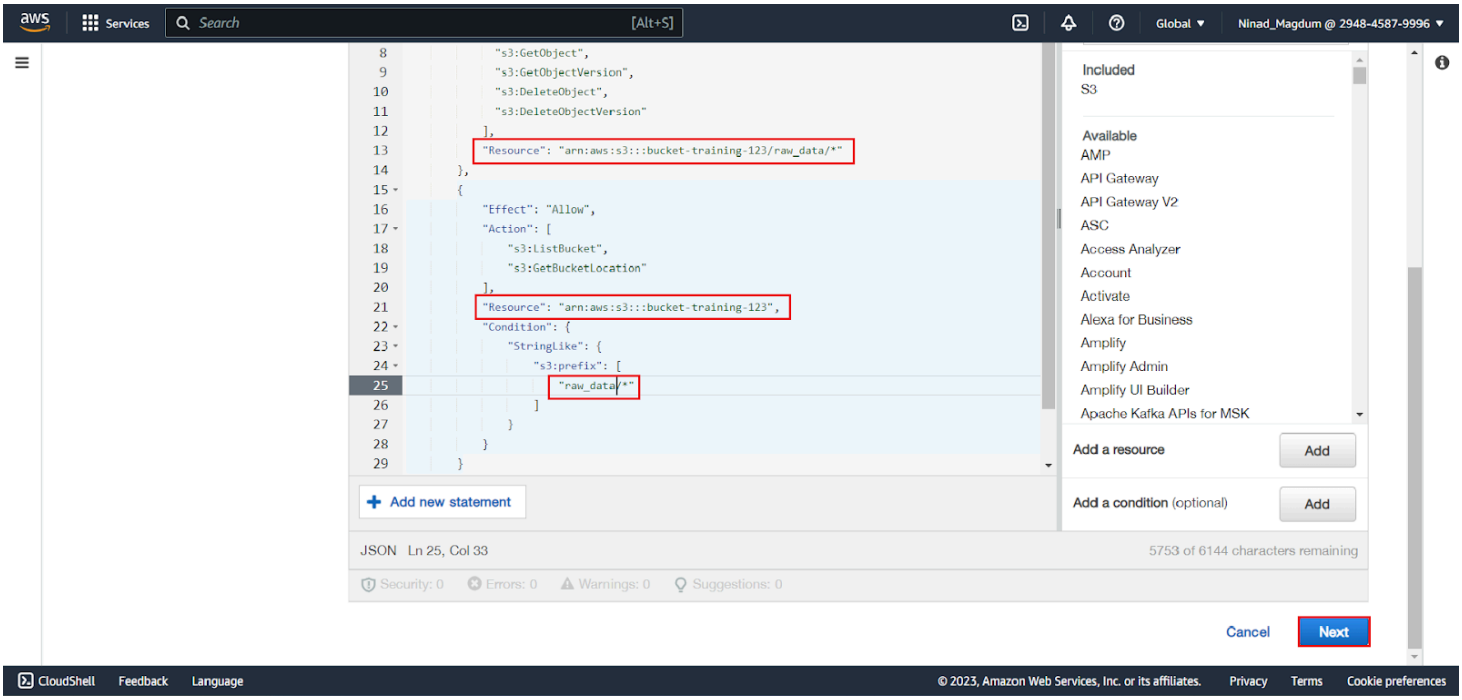
5. Click the JSON tab.



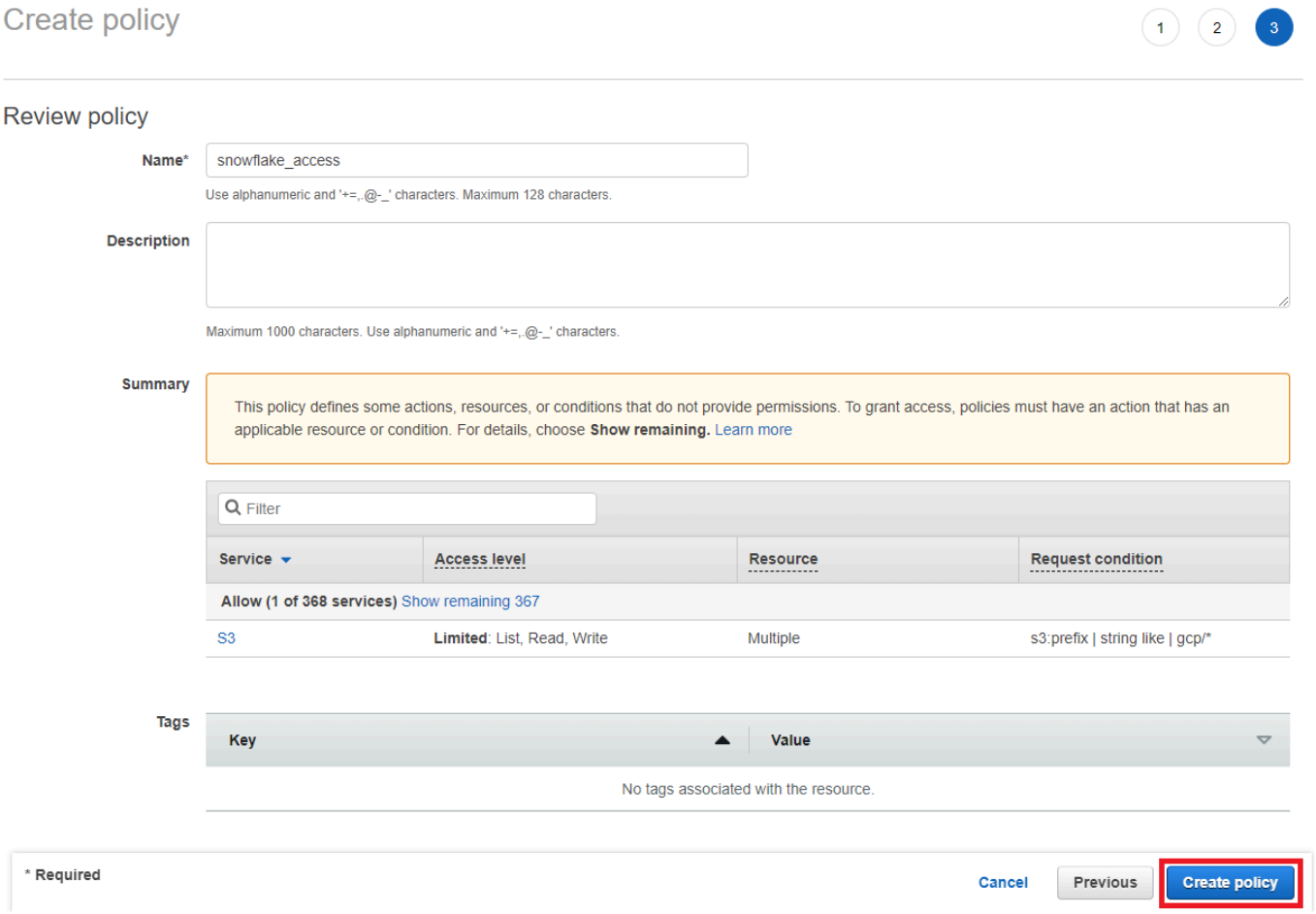
6. Add a policy document that will allow Snowflake to access the S3 bucket and folder. Click on Next. Make sure to replace bucket and prefix with your actual bucket name and folder path prefix.

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
              "s3:PutObject",
              "s3:GetObject",
              "s3:GetObjectVersion",
              "s3:DeleteObject",
              "s3:DeleteObjectVersion"
            ],
            "Resource": "arn:aws:s3:::<bucket>/<prefix>/*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation"
            ],
            "Resource": "arn:aws:s3:::<bucket>",
            "Condition": {
```

```
            "StringLike": {
                "s3:prefix": [
                    "<prefix>/*"
                ]
            }
        }
    }
  ]
}
```

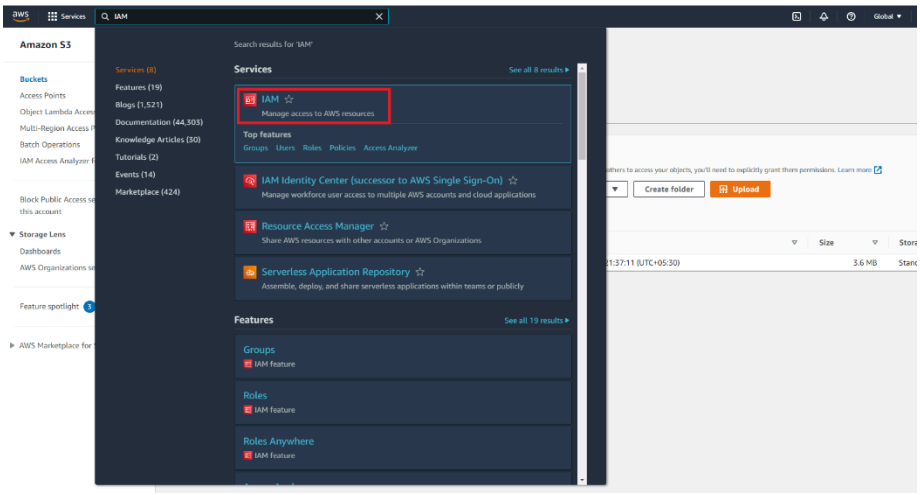


7. Enter the policy name (e.g. snowflake_access) and an optional description. Click Create policy.

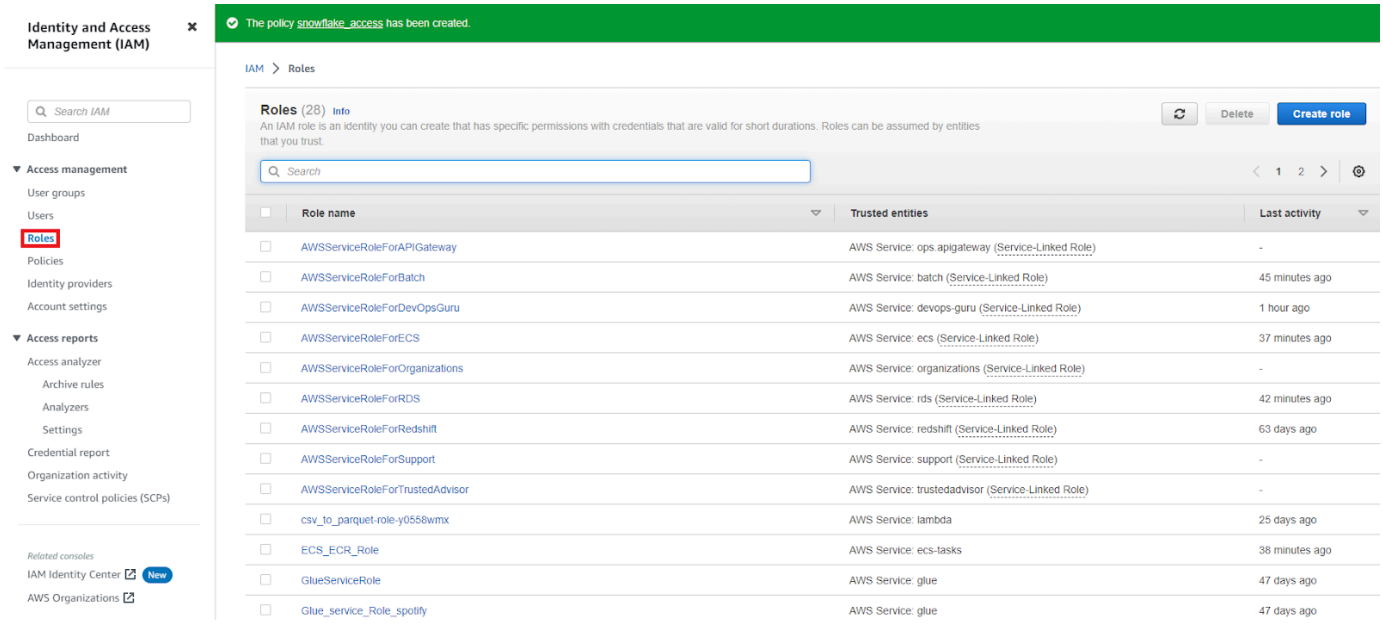


Create IAM Role

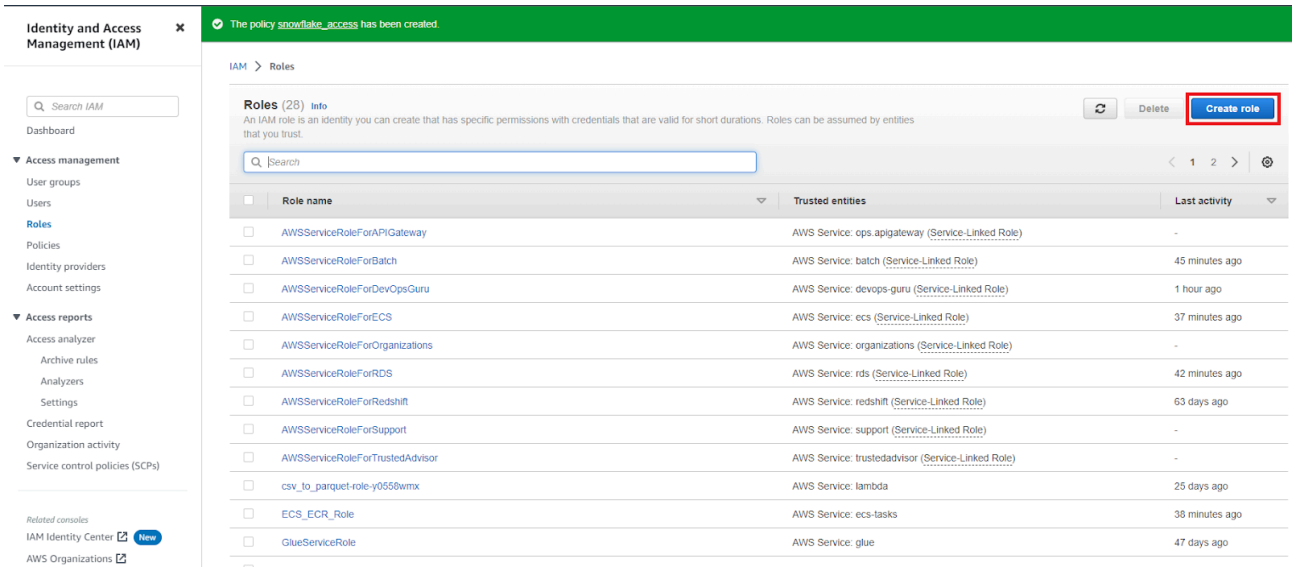
1. Log into the AWS Management Console.
2. From the home dashboard, choose Identity & Access Management (IAM).



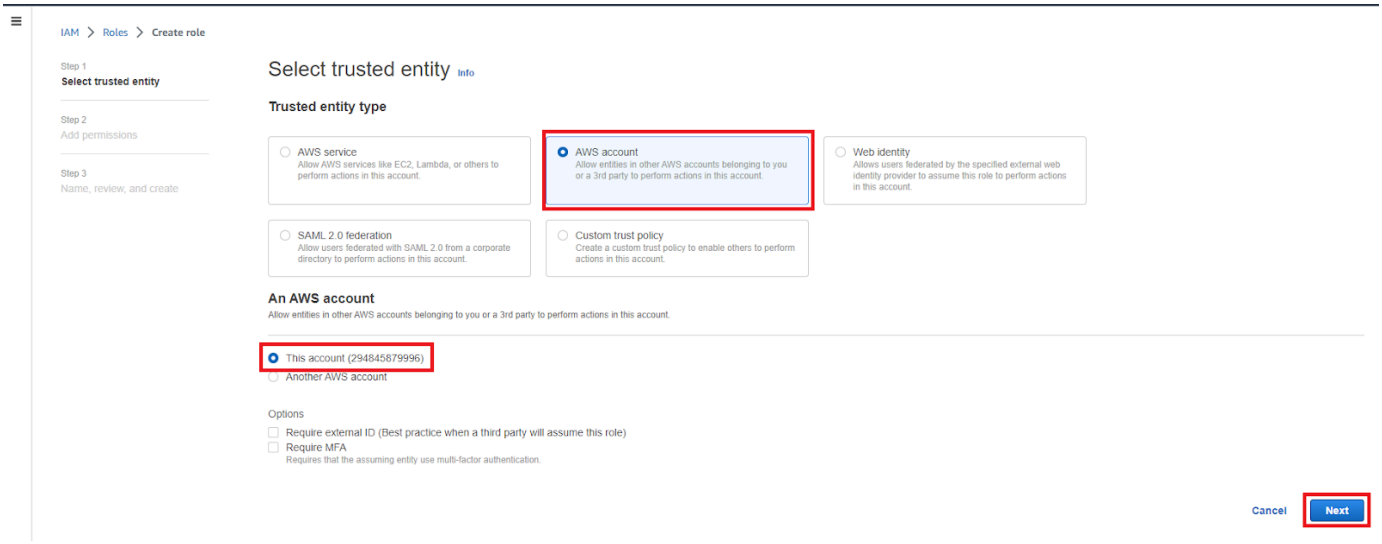
3. Choose Roles from the left-hand navigation pane.



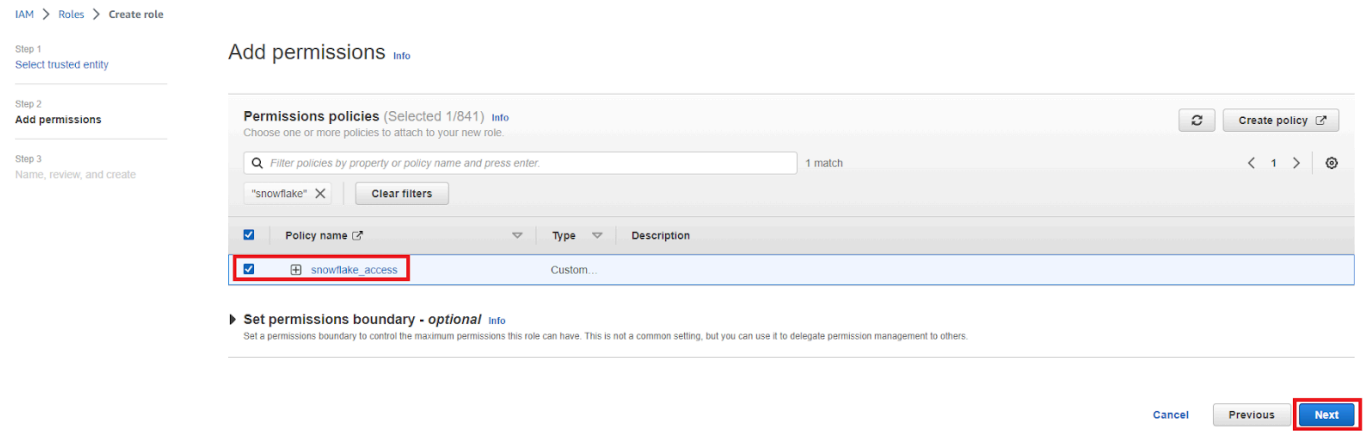
4. Click the Create role button.



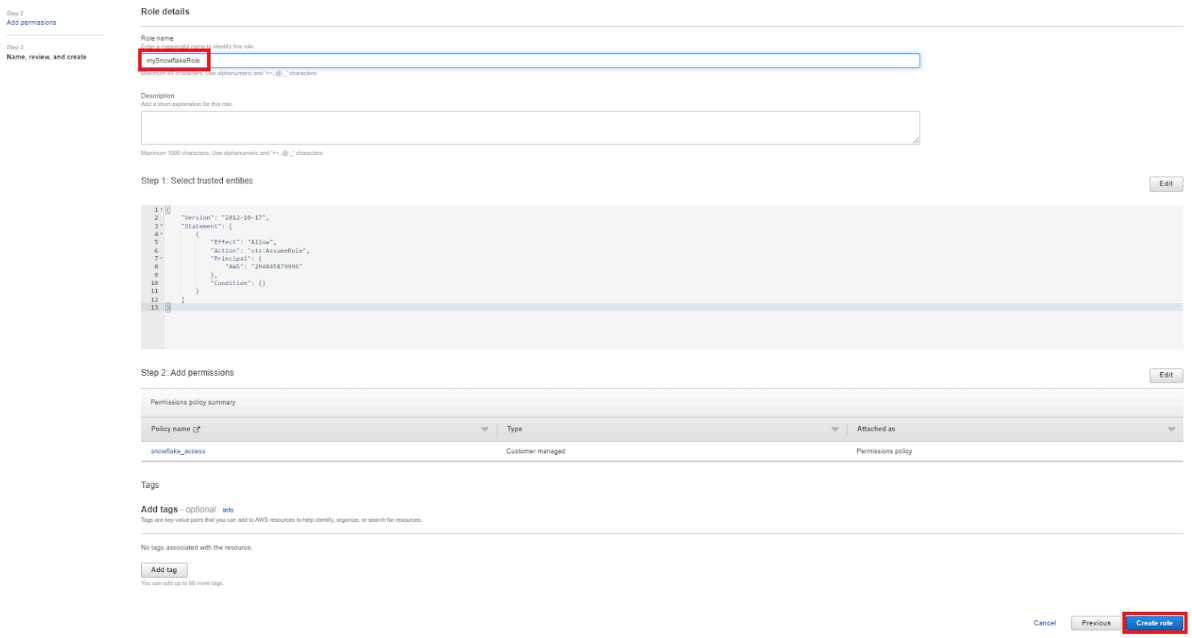
5. Select Another AWS account as the trusted entity type. Click the Next button.



6. Locate the policy you created in previous step for the S3 Bucket (in this topic) and select this policy. Click the Next button.

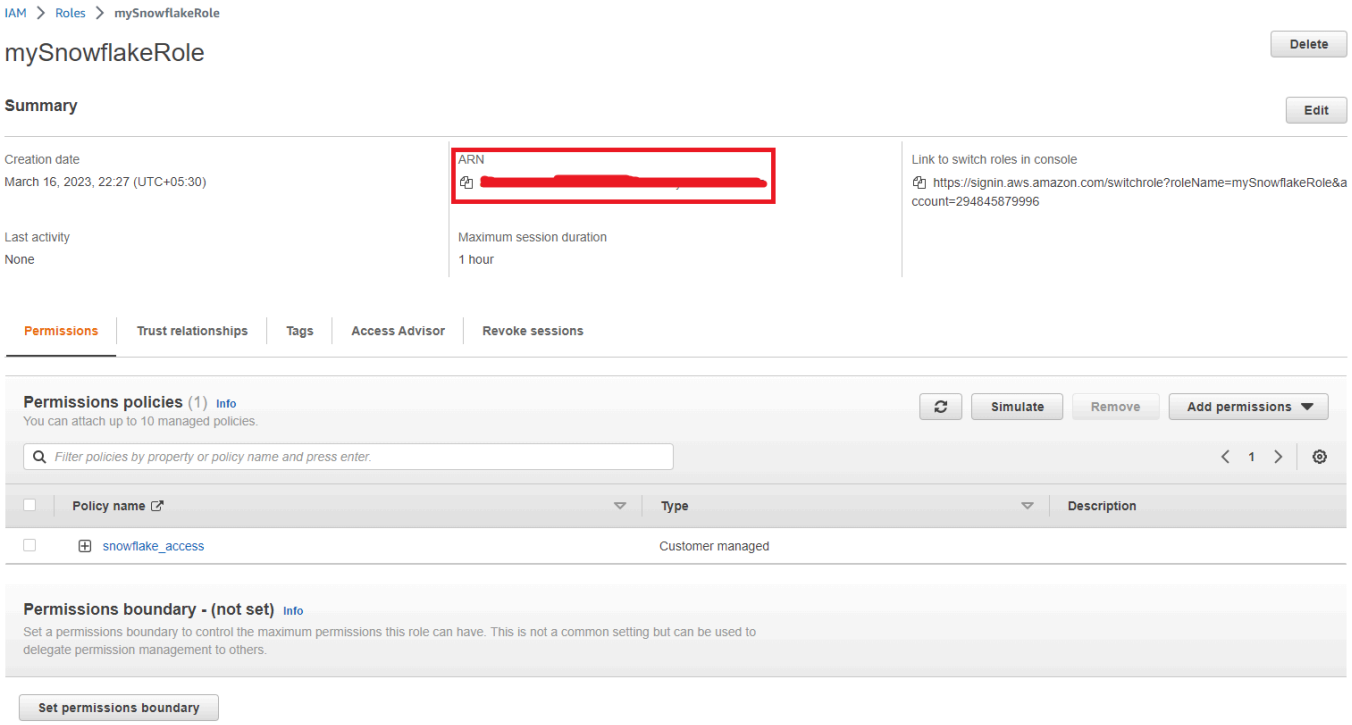


7. Enter a name and description for the role and click the Create role button.



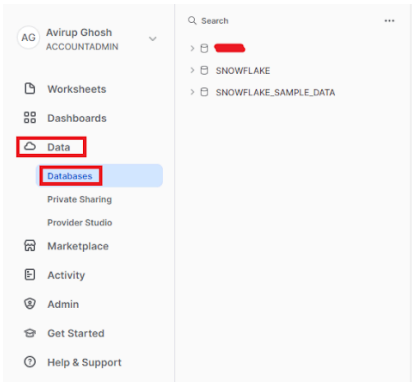
You have now created an IAM policy for a bucket, created an IAM role, and attached the policy to the role.

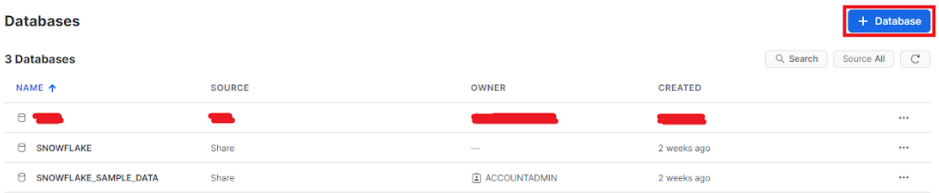
8. Record the Role ARN value located on the role summary page. In the next step, you will create a Snowflake integration that references this role.



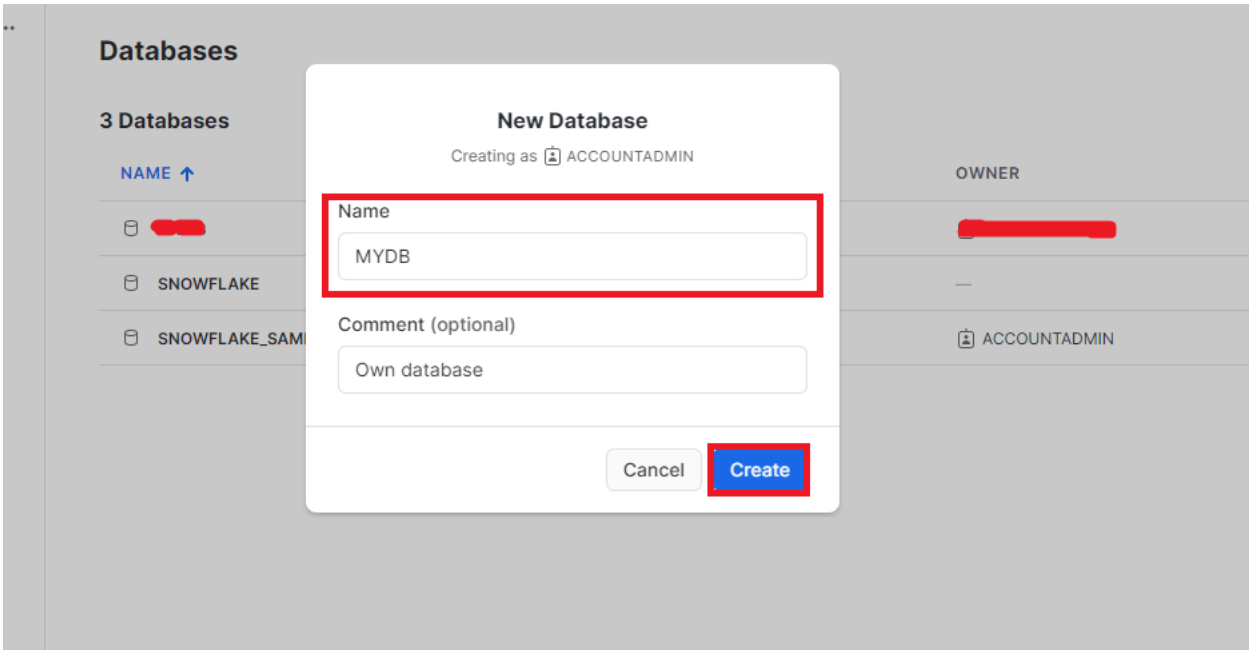
Setting up Cloud Storage Integration

1. Go to Snowflake UI, click on Data and then you have all your available databases. On the right top corner click on Database option to create a new database.

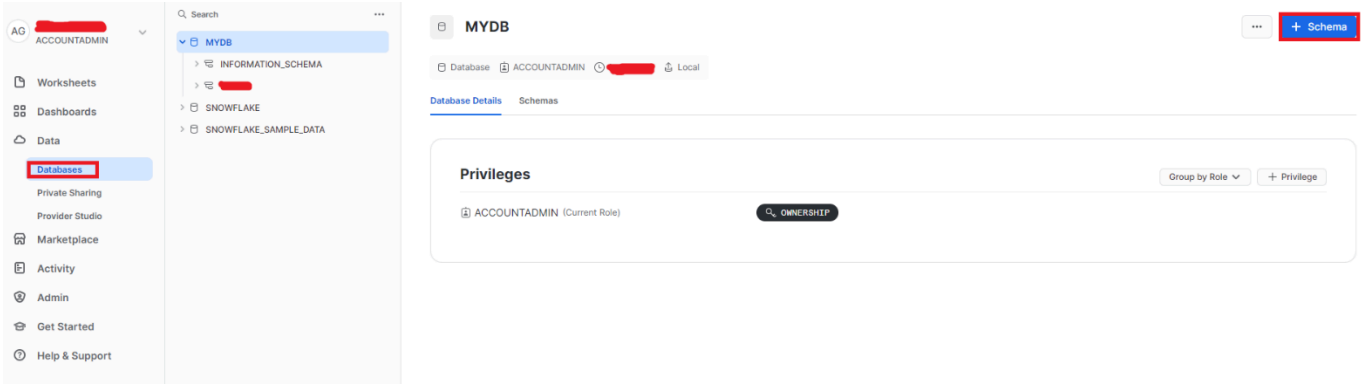




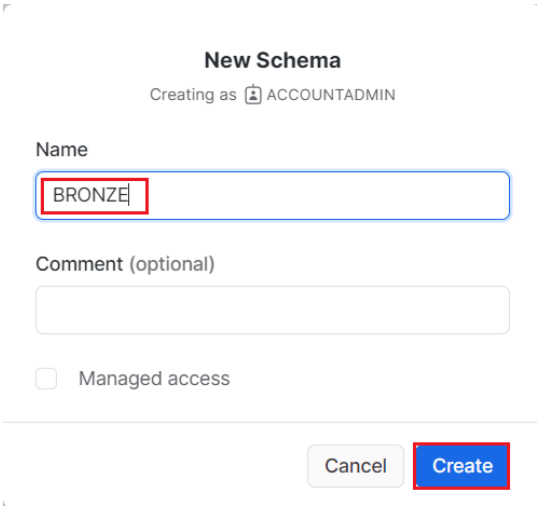
2. Give it a name like MYDB and add some comments. Click on Create to proceed.



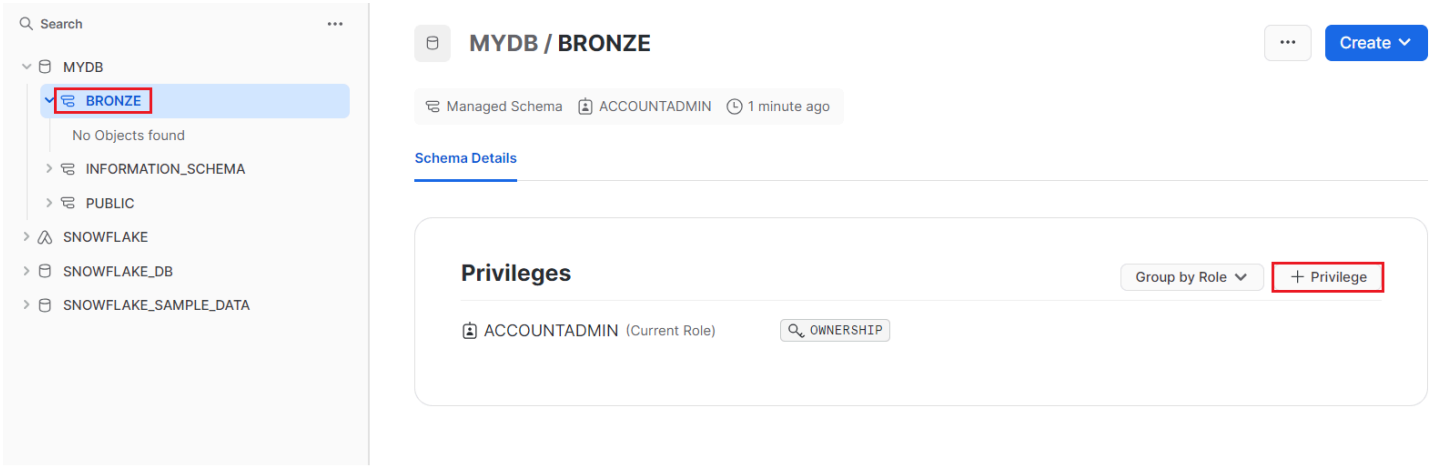
3. Now under databases you can see your newly created database. Click on it and then on the right top corner click on option Schema.



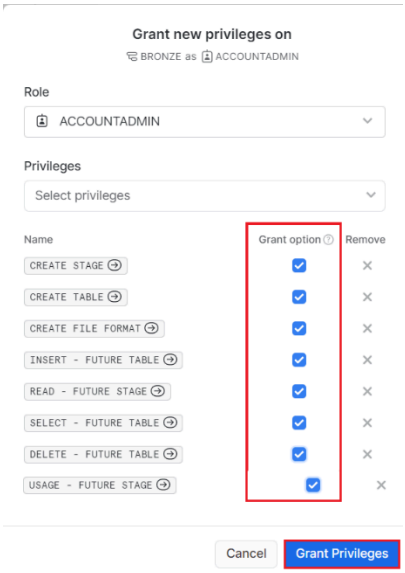
4. Now give it a name like BRONZE. Then click on Create to proceed.



5. Now from the list of schemas under your database select BRONZE. On the right side click on Privilege to add permissions to this schema.

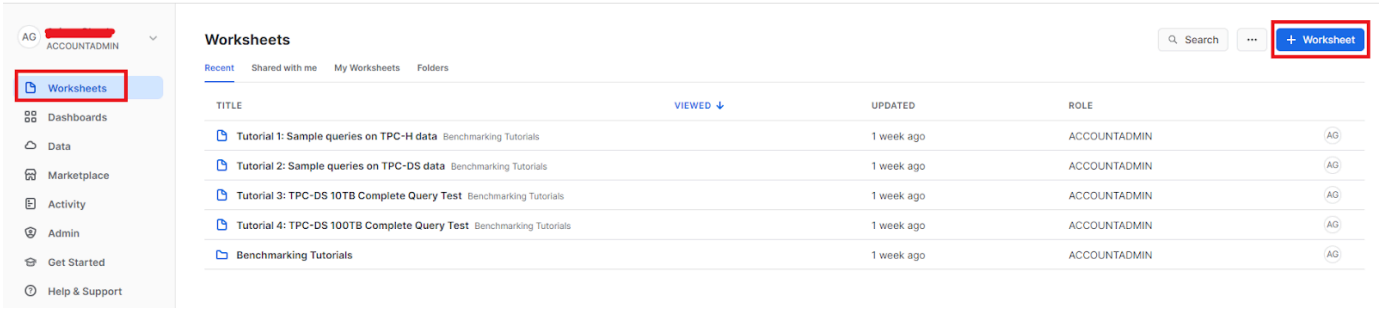


6. Now select role as ACCOUNTADMIN and select the privileges one by one like below. Check Grant option boxes for all privileges and click Grant Privileges.



Now you have privileges to operate on tables and stages.

7. Go to Snowflake worksheets using your account. Then on the left side click on Worksheets and select + Worksheet to open a new SQL worksheet.



8. New worksheet opens. Now run below commands by replacing values in <>:

```
USE SCHEMA MYDB.BRONZE;

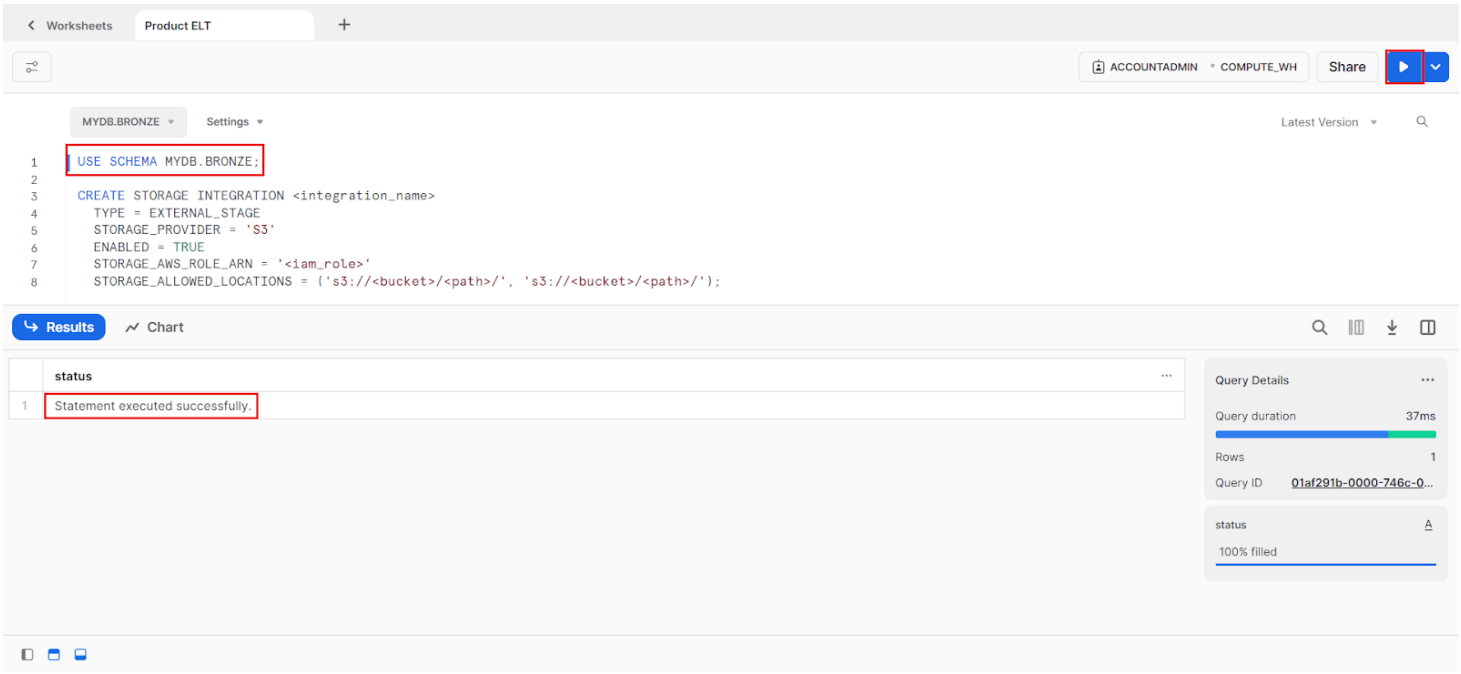

CREATE STORAGE INTEGRATION <integration_name>
  TYPE = EXTERNAL_STAGE
  STORAGE_PROVIDER = 'S3'
  ENABLED = TRUE
```

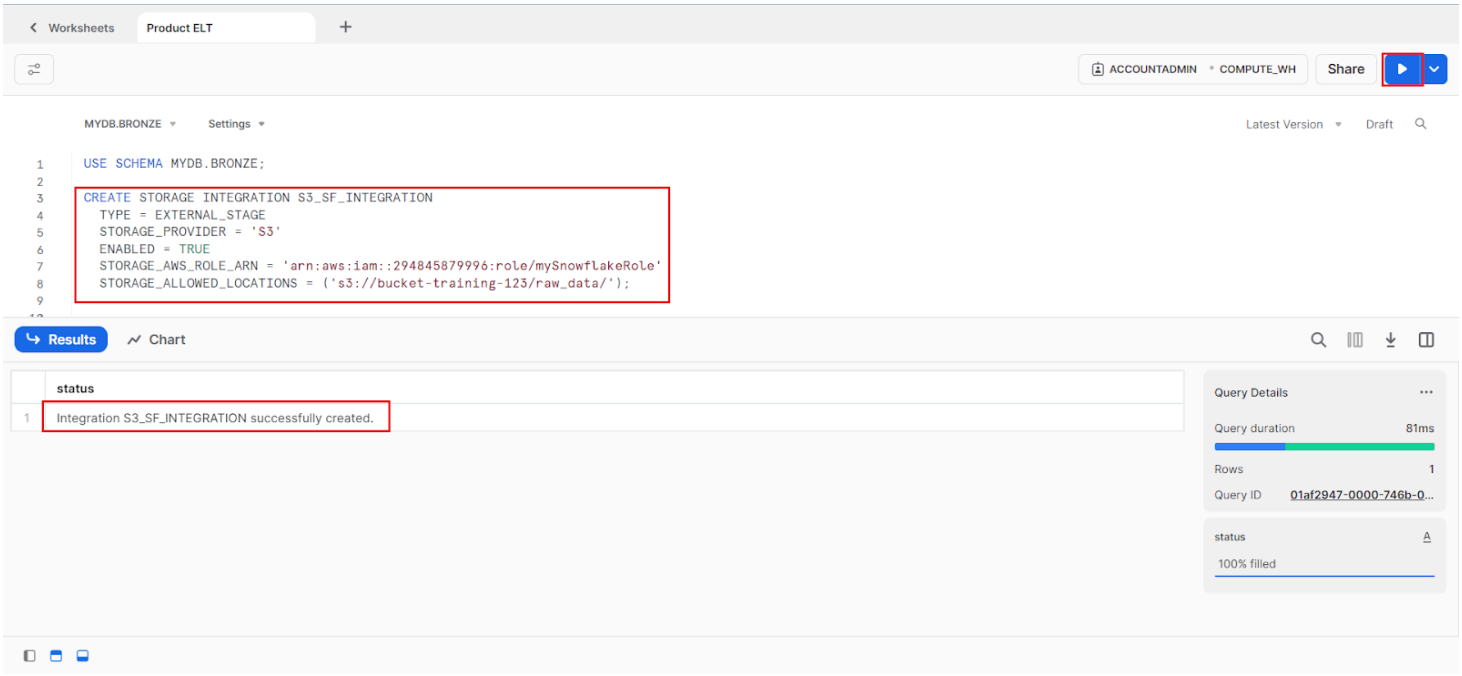
```
STORAGE_AWS_ROLE_ARN = '<iam_role>'
STORAGE_ALLOWED_LOCATIONS = ('s3://<bucket>/<path>/', 's3://<bucket>/<path>/');
```

<iam_role> specifies the Amazon Resource Name (ARN) of the AWS IAM role you create using above steps.

<s3://bucket/path/> specifies the AWS S3 Bucket folder path for the integration to point.

<integration_name> specifies the name for the integration.

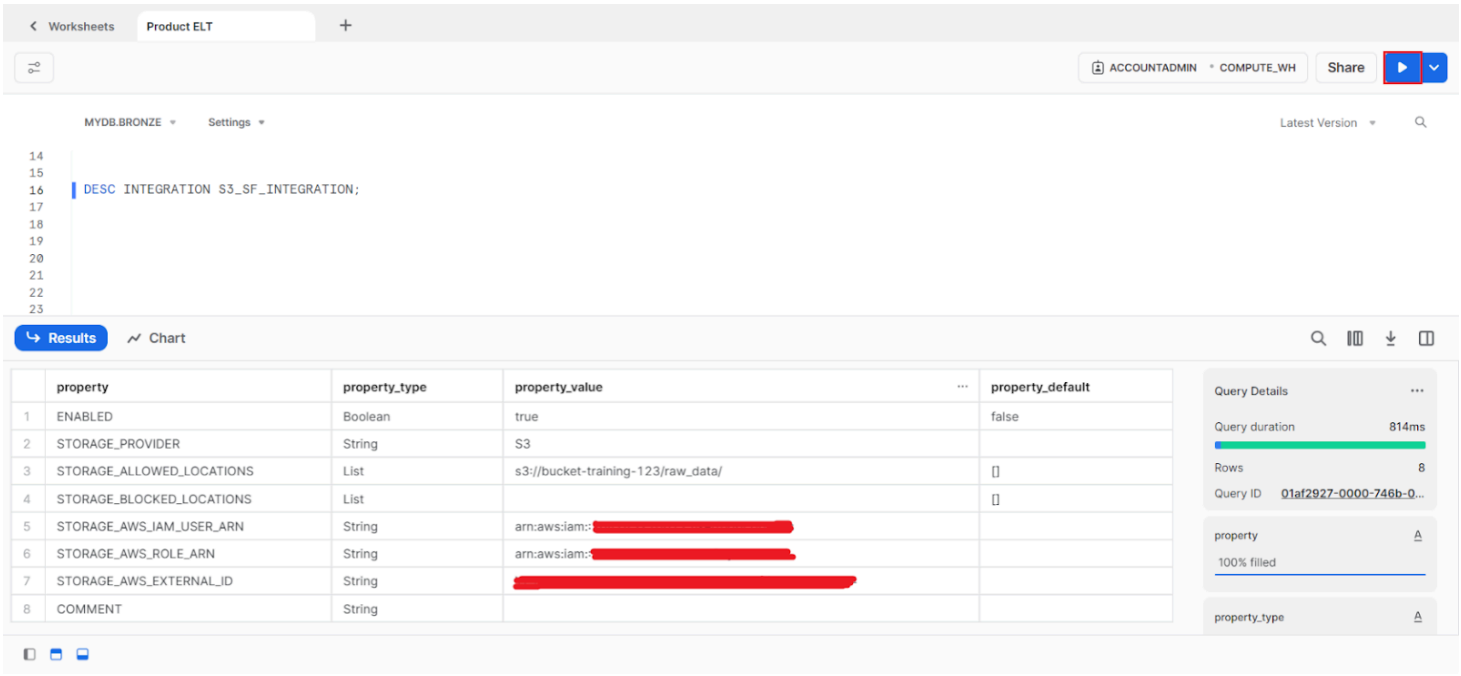




Retrieve the AWS IAM User for your Snowflake Account

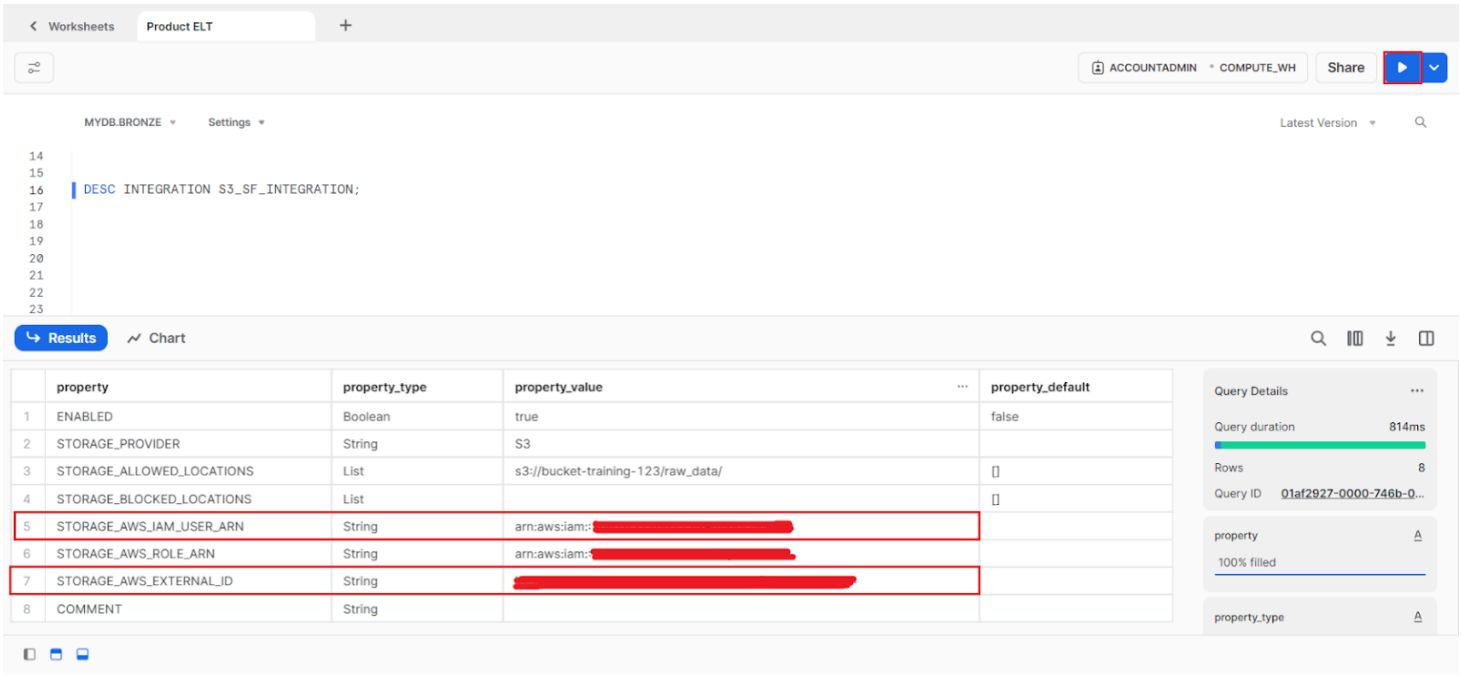
1. Execute the DESCRIBE INTEGRATION command to retrieve the ARN for the AWS IAM user that was created automatically for your Snowflake account.

DESC INTEGRATION <integration_name>;



	property	property_type	property_value	property_default
1	ENABLED	Boolean	true	false
2	STORAGE_PROVIDER	String	S3	
3	STORAGE_ALLOWED_LOCATIONS	List	s3://bucket-training-123/raw_data/	[]
4	STORAGE_BLOCKED_LOCATIONS	List		[]
5	STORAGE_AWS_IAM_USER_ARN	String	arn:aws:iam::	
6	STORAGE_AWS_ROLE_ARN	String	arn:aws:iam::	
7	STORAGE_AWS_EXTERNAL_ID	String		
8	COMMENT	String		

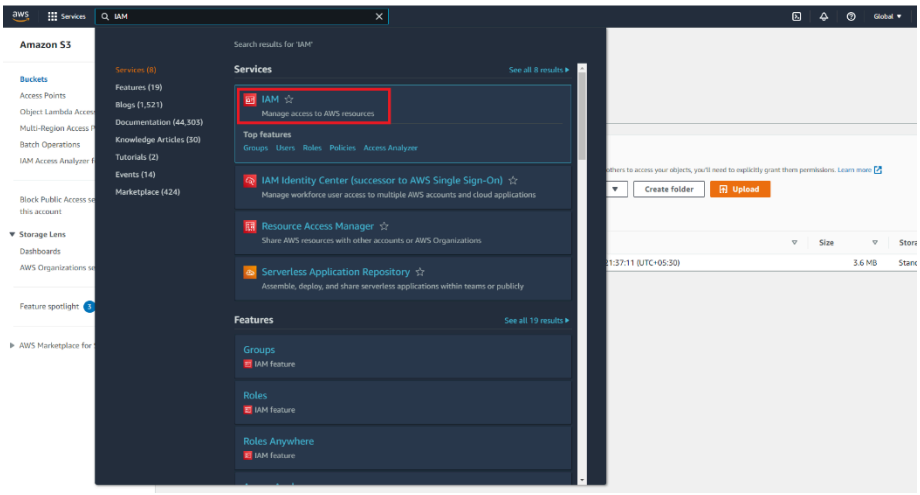
2. Record the following values highlighted with box:



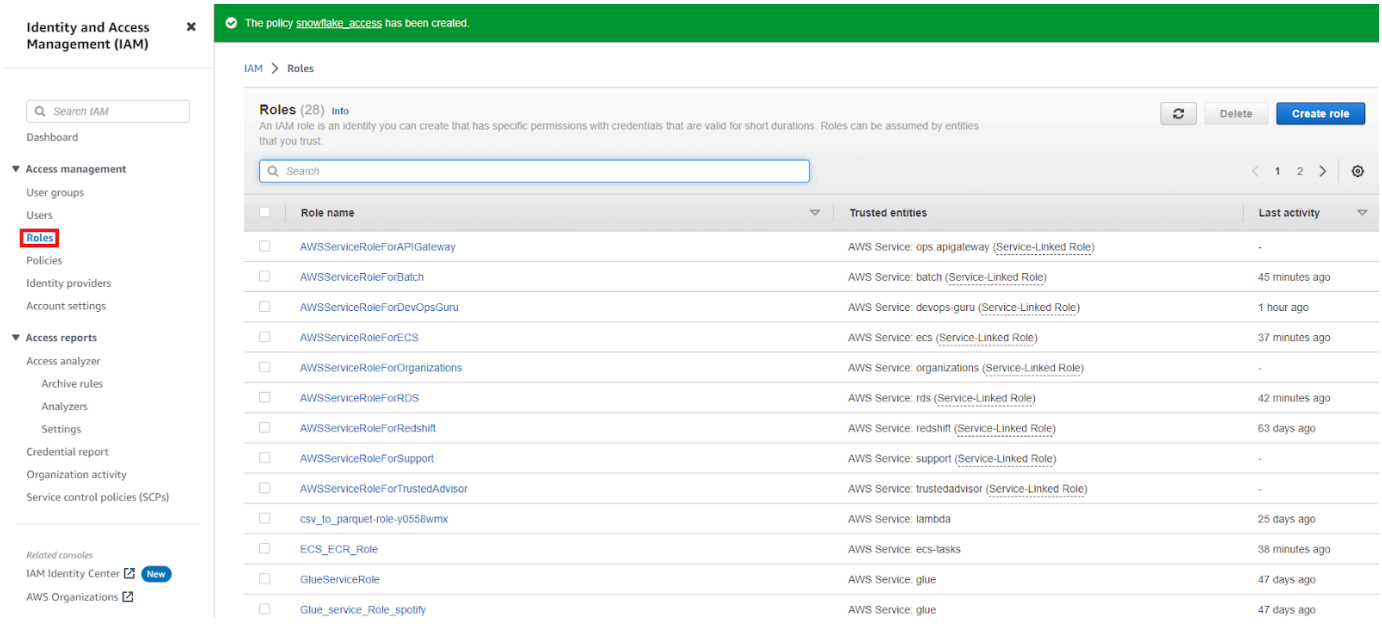
	property	property_type	property_value	property_default
1	ENABLED	Boolean	true	false
2	STORAGE_PROVIDER	String	S3	
3	STORAGE_ALLOWED_LOCATIONS	List	s3://bucket-training-123/raw_data/	[]
4	STORAGE_BLOCKED_LOCATIONS	List		[]
5	STORAGE_AWS_IAM_USER_ARN	String	arn:aws:iam::	
6	STORAGE_AWS_ROLE_ARN	String	arn:aws:iam::	
7	STORAGE_AWS_EXTERNAL_ID	String		
8	COMMENT	String		

Grant the IAM User Permissions to Access Bucket Objects

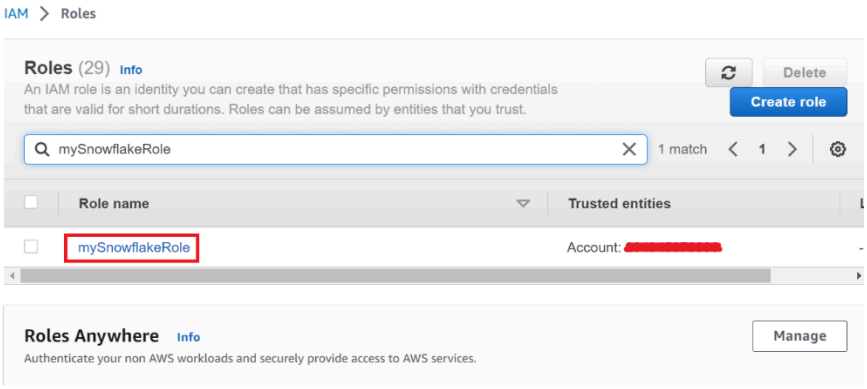
1. Log into the AWS Management Console.
2. From the home dashboard, choose Identity & Access Management (IAM).



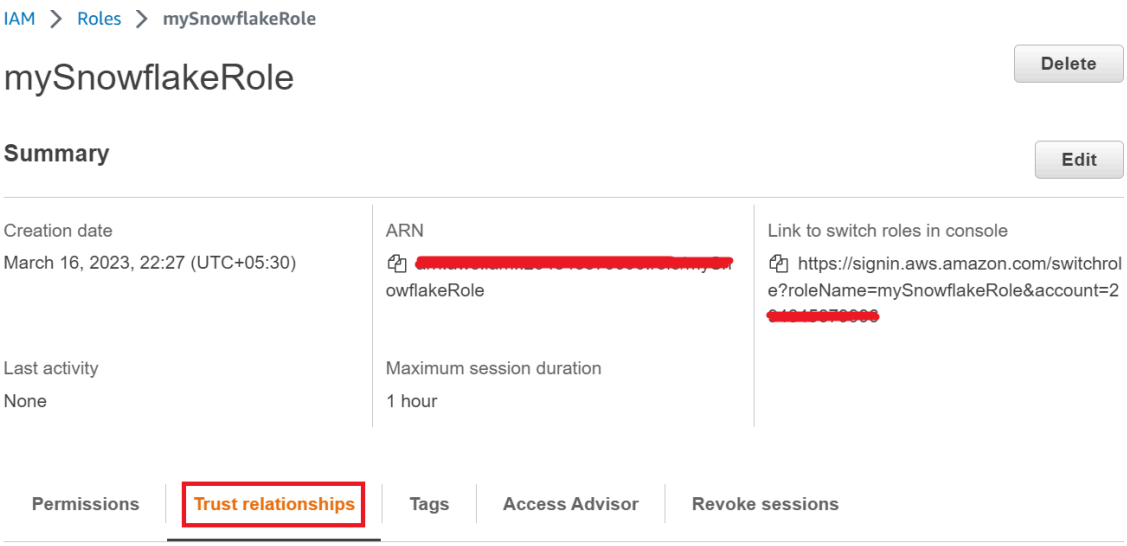
3. Choose Roles from the left-hand navigation pane.



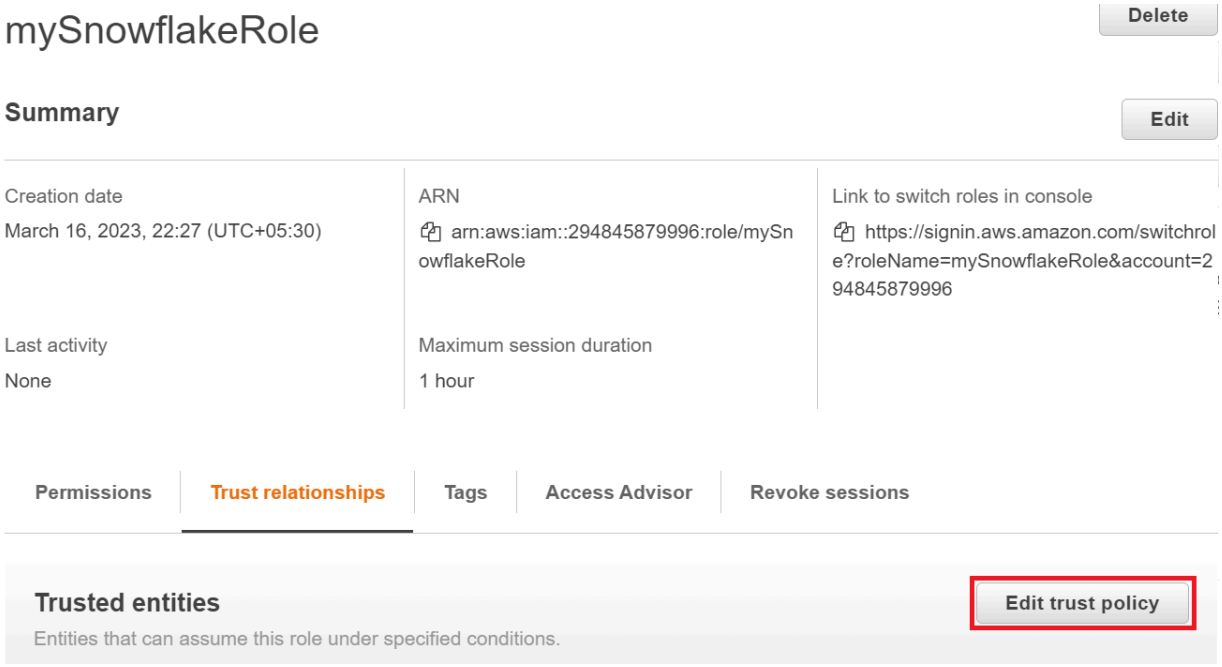
4. Click on the role you created in the process Create the IAM Role in AWS (in this topic).



5. Click on the Trust relationships tab.



6. Click the Edit trust relationship button.



7. Modify the policy document with the DESC STORAGE INTEGRATION output values you recorded in process Retrieve the AWS IAM User for your Snowflake Account (in this topic):

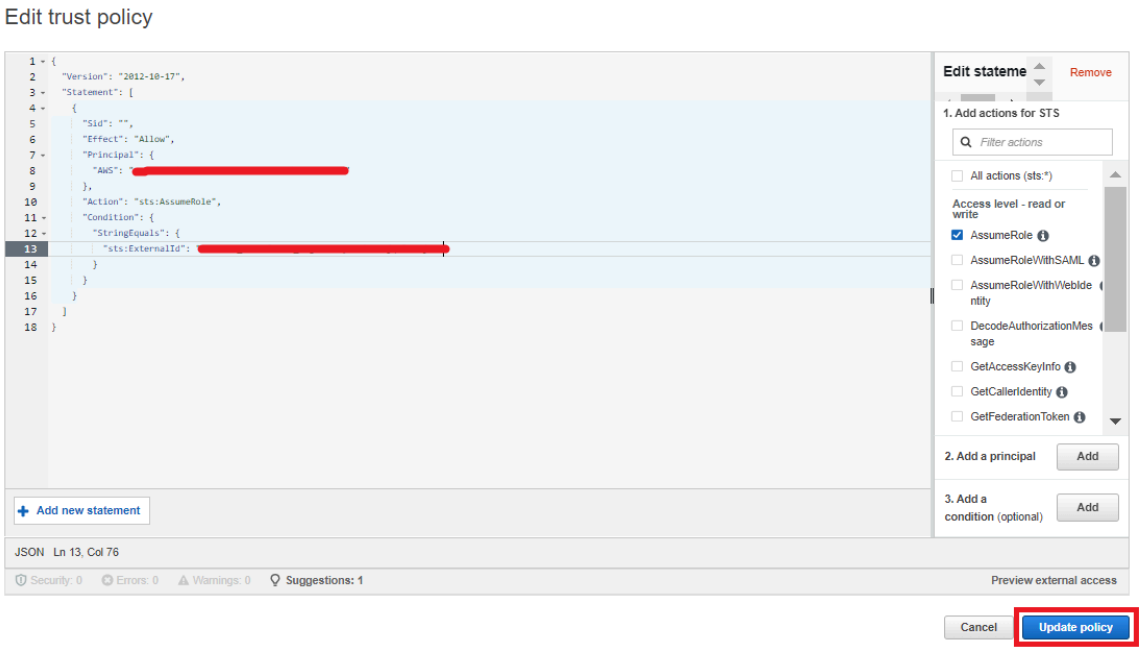
<snowflake_user_arn> is the STORAGE_AWS_IAM_USER_ARN value you recorded.

<snowflake_external_id> is the STORAGE_AWS_EXTERNAL_ID value you recorded.

```
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Sid": "",
   "Effect": "Allow",
   "Principal": {
    "AWS": "<snowflake_user_arn>"
   },
   "Action": "sts:AssumeRole",
   "Condition": {
    "StringEquals": {
     "sts:ExternalId": "<snowflake_external_id>"
```

```
        }
      }
    }
  ]
}
```

8. Click on Update policy.



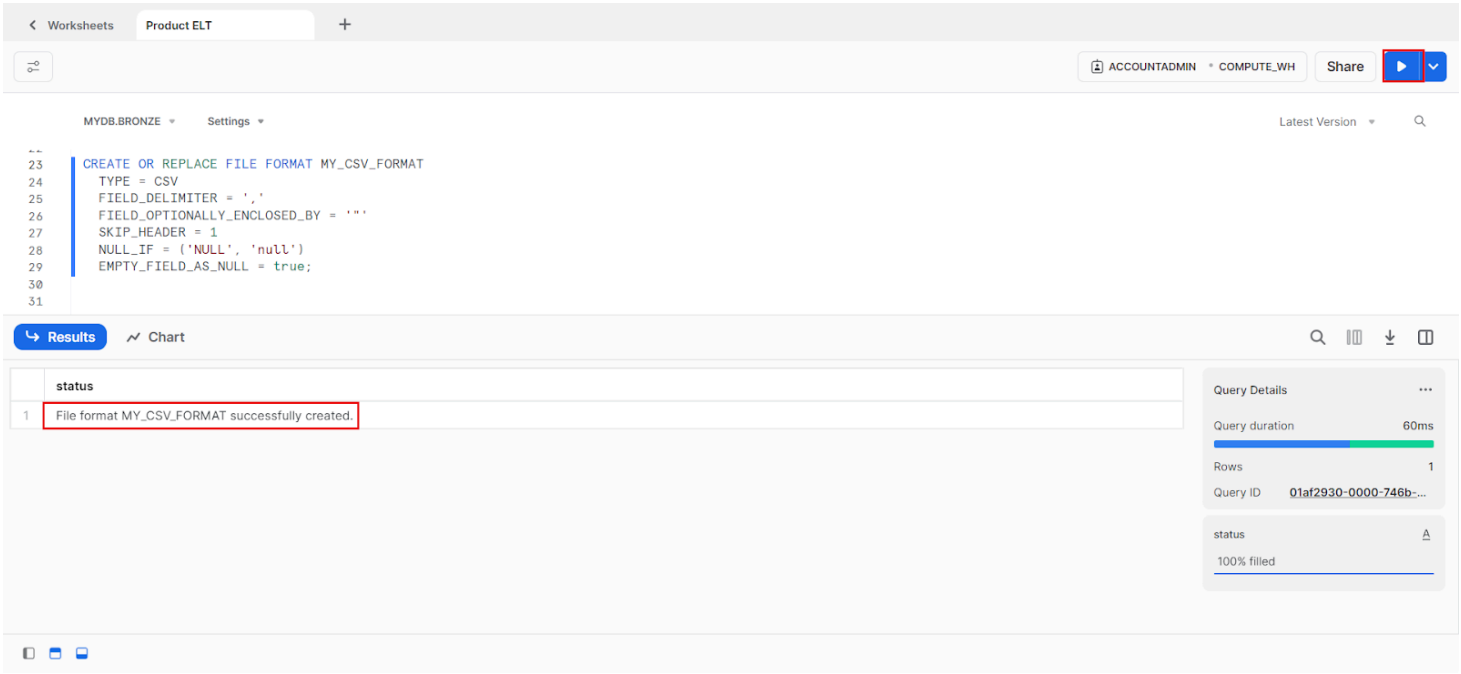
Create an External Stage

1. Create an external (i.e. S3) stage that references the storage integration you created in process Create a Cloud Storage Integration in Snowflake (in this topic).

2. Create a file format (may use any name of your choice) that will be used to read .csv format from the external stage made on top of the AWS S3 bucket folder. Use below command to create the file format by running it.

```
CREATE OR REPLACE FILE FORMAT MY_CSV_FORMAT
  TYPE = CSV
  FIELD_DELIMITER = ','
  FIELD_OPTIONALLY_ENCLOSED_BY = '"'
  SKIP_HEADER = 1
  NULL_IF = ('NULL', 'null')
```

EMPTY_FIELD_AS_NULL = true;



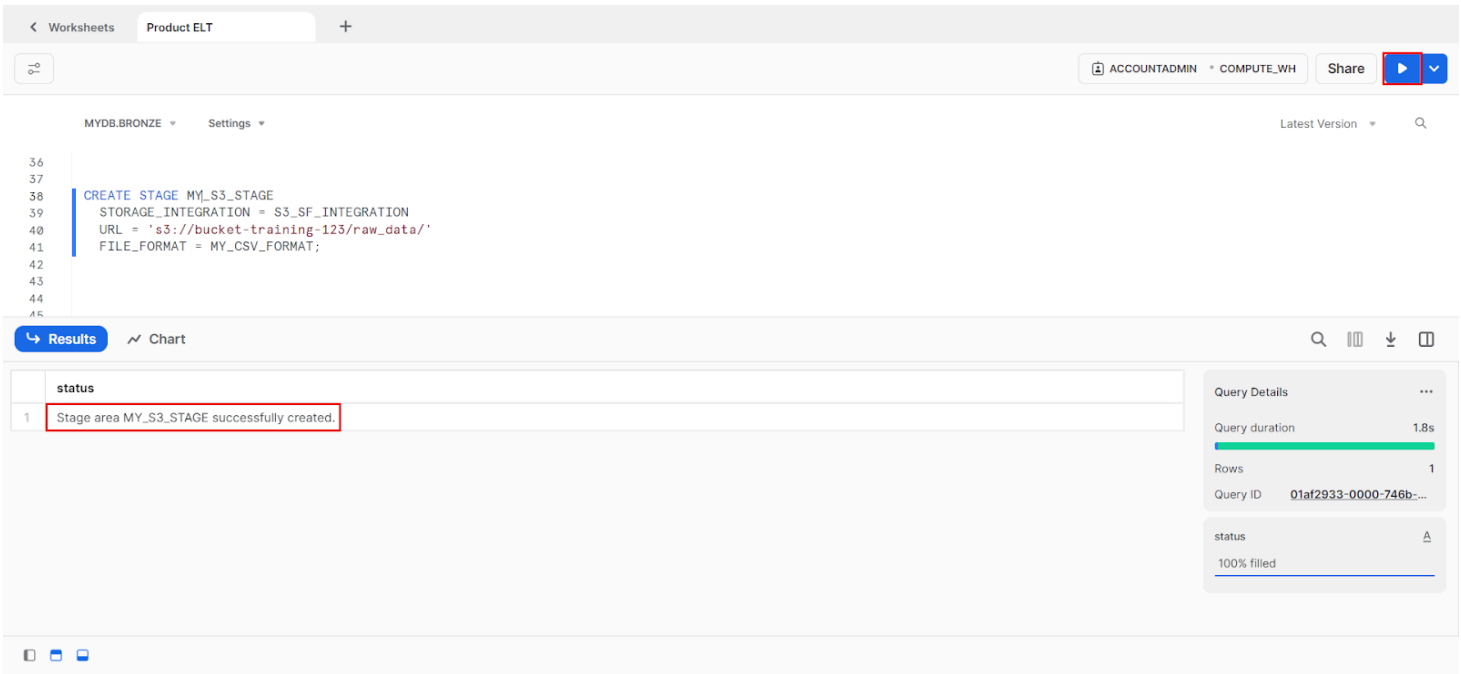
3. After run below commands on worksheet. Run the below query by changing your bucket path & format:

CREATE STAGE <stage_name>

STORAGE_INTEGRATION = <integration_name>

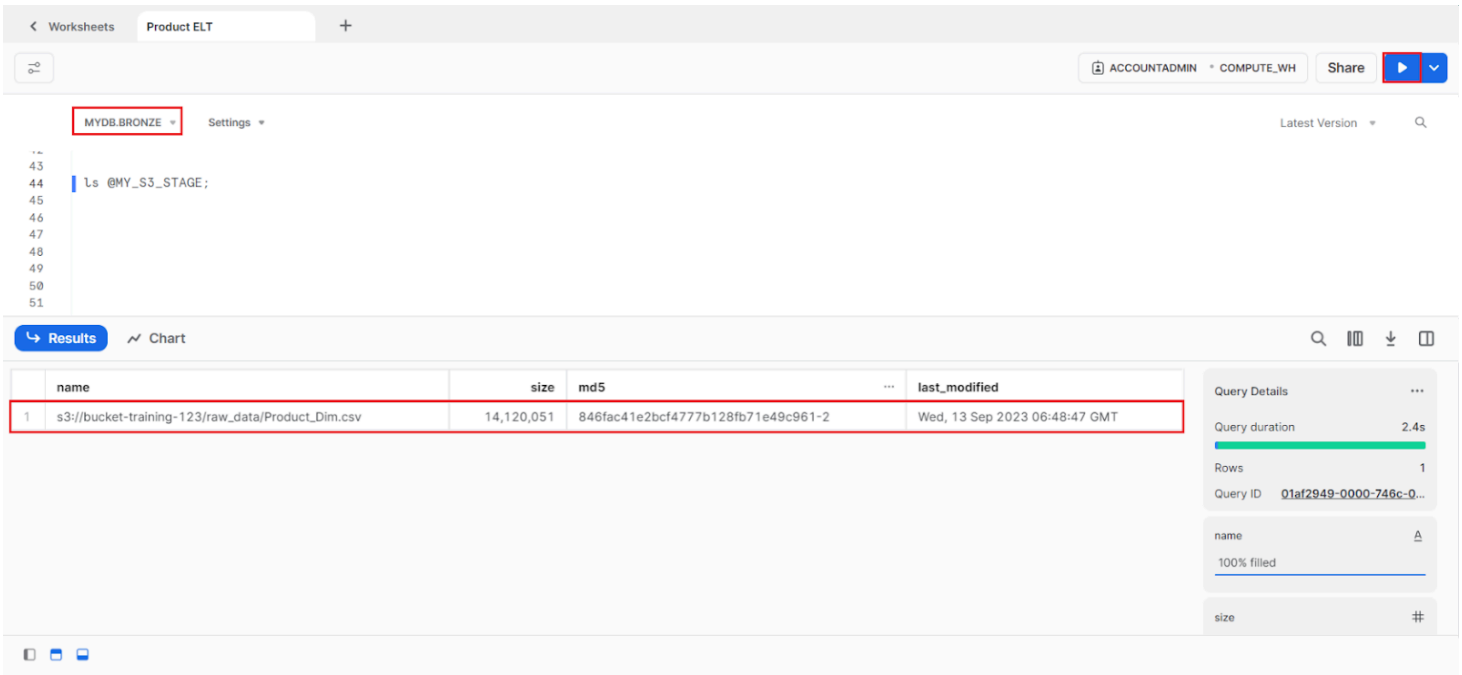
URL = '<bucket_path1>'

FILE_FORMAT = <my_format>;



Test your stage using below command to see files in AWS S3 bucket like below.

ls @<stage_name>;



Successfully setup done of S3 with Snowflake using External Stage.

IMPORTANT: Legal Disclaimers

*Data Education Holdings LLC is not responsible for any costs incurred while completing this lesson. Data Education Holdings LLC is not part of Amazon.com, Inc. It is also not endorsed by Amazon, AWS, or any of the affiliated subsidiaries.

© Data Education Holdings LLC